Tension Resolution and Sustaining Knowledge Flows in Online Communities

Completed Research Paper

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Abstract

Online communities bring together members of common interest and are recognized as generators of significant knowledge. In healthcare, online communities are a medium for patients to share their stories, gain peer support, and learn more about medical conditions. To succeed, these communities need to sustain their knowledge flows and resolve arising tensions. The current literature suggests that participation dynamics and emergent organizational structure are sufficient to resolve tensions in online communities. Based on increased evidence of the need of formal management to sustain online communities, we propose that emergent structure and roles may not always be sufficient when tensions are prevalent and stakes are high such as in online patient communities. We conduct an inductive field study of Mayo Clinic Connect, a leading online patient community. The findings suggest that formal management and emergent leadership play a complementary role to resolve tensions and sustain knowledge flows. We theorize a virtuous cycle of moderation and socialization that effectively integrates the effort of the community formal management and emergent leaders.

Keywords: Healthcare, online communities, tensions, moderation, socialization, knowledge flows

Introduction

Online communities bring together members who share common interests and contribute valuable knowledge and expertise. Today, online communities are recognized as generators of significant knowledge (Faraj et al. 2016). Examples of community-based knowledge work include collaborative online encyclopedias such as Wikipedia, digital artifact design communities such as remixing communities and open-source communities, and expert question-answers communities. In healthcare, online communities such as PatientsLikeMe, HealthUnlocked, and Mayo Connect have become a medium for patients to share their stories, gain peer support and search for medical information (Yan et al. 2015). Seventy-two percent of internet users have looked at health information online within the past year. Thus, online patient communities are becoming forums for patients to share their stories, gain peer support, and search for medical information (Goh et al. 2016; Yan et al. 2015). Membership in patient communities is associated with better health, behavior and medical knowledge (Goh et al. 2016).

The extant literature recognizes the voluntary nature of the contribution and the emergent nature of organizing work in online communities (Faraj et al. 2011, 2016). Members of online communities are motivated to work together and help each other (Sproull et al. 2005). Over time, members in online communities build shared understanding, social norms, and trust (Ren et al. 2012). Online communities also evolve elaborate organizational structures (Butler et al. 2008). Some members become leaders and take an active role in helping and guiding other members (Johnson et al. 2015; Oh et al. 2016). Members' sociality and emergent organizational structure make online communities an effective space for knowledge flows (Faraj et al. 2016).

Notwithstanding the sociality of online communities, tensions among members arise during collaborative knowledge work. Some of these tensions center around knowledge co-production and the desire of members to shape the knowledge artifact in their way (Kane et al. 2014). Other forms of tensions arise because of the open-ended nature of membership – bringing together members of different backgrounds and often unclear goals and expectations (Faraj et al. 2011). Additional tensions such as anti-social behavior may create destructive knowledge flows such as spamming, vandalism, and spreading false information that challenge the capacity of online communities to generate valuable knowledge while remaining open and inclusive (Halfaker and Taraborelli 2015).In healthcare communities, tensions are more prevalent because of the sensitivity of medical and personal information (McKee 2013) and the quality, accuracy, and trustworthiness of medical information that is shared online (Benigeri and Pluye 2003; Marrie et al. 2013; Ogah and Wassersug 2013).

To succeed, online communities need to resolve rising tensions. Two primary perspectives on how online communities resolve tensions exist in the literature: organizational structure and emergent dynamics (Kane et al. 2014). The organizational structure perspective emphasizes the importance of a core group in the community to take active leadership roles for resolving these tensions (Butler et al. 2008; O'Mahony and Ferraro 2007). The emergent dynamics perspective suggests that online communities can improvise generative responses to these tensions when they arise (Faraj et al. 2011; Shaikh and Vaast 2016).

Both perspectives agree on the emergent nature of the organization structure and generative responses. They suggest that online communities can resolve tensions in a bottom-up fashion ruling out the need for formal management or planned resolution protocols. However, there is increased recognition of the role of active management for sustaining online communities (Barrett et al. 2016). Today, many thriving online communities employ professional management that plans participation guidelines and moderation protocols as well as monitors developing needs of the community (Kumar et al. 2017).

In this paper, we are interested in understanding the role of formal online community management in resolving tensions which results in sustaining knowledge flows. We seek to understand how the community management of an online community can complement the emergent structure and participation dynamics. With the increase of formally managed communities, especially in health care fields, no study has yet to determine if formal moderation complements emergent sociality to resolve tensions. To gain a deep understanding of this phenomenon, we conduct an inductive field study on Mayo Clinic Connect (connect.mayoclinic.org), a leading online patient community. Using grounded theory methodology, we

collect observational and interview data from community managers, emergent leaders and other key members over three years.

Our findings indicate that in addition to socialization as a key mechanism to reduce tensions, planned top-down moderation practices play a significant role as well. These practices serve as control mechanisms that are employed collaboratively by the community management and its emergent leaders. The implementation of these control mechanisms leverages the emergent nature of roles in the community. We theorize how two processes that we label "informalizing the formal" and "formalizing the informal" serve to integrate the top-down and bottom-up participation dynamics. Formal management can formalize informal practices through processes such as selection, training, and emulation. At the same time, socialization may informalize formal management roles and policies through new emergent social norms. We develop a framework that depicts a virtuous cycle of socialization and moderation that can be used to resolve tensions sustain knowledge flows in online communities.

In the backdrop of prior work emphasizing the emergent structure and generative dynamics in online communities, our work highlights the importance of formal control mechanisms in resolving tensions and sustaining knowledge flows. Our inductive theory shows how the virtuous cycle of moderation and socialization reduces tensions and sustains productive knowledge flows especially in contexts where pro-social behavior stakes are high such as in healthcare communities. Our work shows how formal management and emergent leadership can complement each other and promote community success. We further suggest a more nuanced understanding of online communities not only as emergent organizational structures but also as *organized emergent structures* in which formal control and socialization can be complements.

Tensions in Online Communities

An online community is a group of people who share a keen common interest, form relationships and interact dominantly online (Sproull and Arriaga 2012). Knowledge collaboration is the distinctive outcome of online communities that differentiates them from other online mediate communication forms (Faraj et al. 2016). Online communities are capable of creating precious knowledge (Benkler et al. 2015). For example, the economic value of Wikipedia is estimated at $6.6 billion. Similarly, the development cost of the Linux operating system is about $10.8 billion (McPherson et al. 2008). Members of the Stack Exchange community contribute answers for 140,000 questions per month (Posnett et al. 2012).

Prior work thoroughly examined what makes online communities an effective space for knowledge creation. This work recently converged on characterizing online communities as emergent forms of organization that can operate and is effective without formal structural mechanisms (Zammuto et al. 2007). Knowledge collaboration in online communities can occur "without the structural mechanisms traditionally associated with knowledge collaboration in organizational teams" (Faraj et al. 2011, p. 1225). Emergent sociality unfolds as members participate and substitute formal organizational structure including stable membership, goal setting, and knowledge transfer (Faraj et al. 2016).

Defined as "the tendency of people to seek related others, build social ties, and organically assemble" (Faraj et al. 2016, p. 4), sociality drives the productive knowledge flows in online communities. The motivation to contribute knowledge stems from members' prosocial behavior to help others (Sproull et al. 2005). Repeated interactions among members lead to developing shared understanding, prevailing social norms, and trust (Ren et al. 2012). Repeated interactions also lead members to identify others as leaders (Faraj et al. 2015; Johnson et al. 2015). Community leaders play an essential role in promoting and sustaining the participation of other members (Oh et al. 2016).

Notwithstanding the prosocial behavior of members and emergent sociality in online communities, tensions arise in many situations. The Oxford English Dictionary defines tension as "A strained state or condition resulting from forces acting in opposition to each other." These forces are often "ideas or qualities with conflicting demands or implications." Such opposing forces are abundant in online communities because of the large number of members and resources involved in contributing and integrating knowledge (Arazy et al. 2016; Faraj et al. 2011). For example, the desire of community members to shape the knowledge artifact in different ways can lead to "edit wars" Kane et al. (2014). This tension arises when one member

changes the content of a page, and another member reverts the content to its original form. Several other tensions arise because of the fluid boundaries of online communities (Faraj et al. 2011). These tensions are the consequences of the flow of resources in and out of the community. These tensions are not necessarily negative. Many often, tensions can result in positive outcomes. It is how the tension is resolved that determines the outcome. Finally, online communities are increasingly dealing with aggressive antisocial behavior that can stress their capacity to sustain knowledge flows (Kumar et al. 2017).

Two primary perspectives on how online communities resolve tensions exist. The first perspective suggests that generative social processes create mechanisms to resolve the tensions, and as the resources change over time, the online community adapts accordingly with new socialized processes to address the new tensions (Faraj et al. 2011). Such generative processes enable members to enact roles in the moment, mobilize resources, and channel participation to resolve the arising tensions. For example, in the Linux Kernel open-source community, elite and junior developers often had different opinions on how to change the codebase. Developers often improvised a "digital fold" by aligning multiple technologies to allow for exclusive codebase changes that are later negotiated with the community (Shaikh and Vaast 2016).

The second perspective emphasizes the importance of a core group of members in the resolution of tensions. When a tension such as an edit war escalates, the leaders of the community step in to resolve (Kane et al. 2014). Although online communities do not usually have a formal structure, they can still emerge organizational structure and leadership roles over time (Butler 2001; O'Mahony and Ferraro 2007). Still, processes for resolving tensions are often missing. Therefore, the community leadership deliberates rather than executes a resolution, a process that results in stabilizing the knowledge artifact as well as shaping members' further contributions to it (Kane et al. 2014).

The two perspectives of organizational structure and emergent dynamics explain how online communities resolve tensions. However, they suffer from two shortcomings. First, they are far from being perfect. The improvised nature of generative responses means they are not guaranteed to happen or be effective when they do (Shaikh and Vaast 2016). When the community leadership makes a deliberation, there is often a lack of clear guidelines which serves to lengthen the time needed to resolve (Kane et al. 2014). Meanwhile, many members become dissatisfied with the process and end up quit contributing to the artifact or leaving the community (Arazy et al. 2016). Second, the two perspectives are focused on knowledge-flow tensions but remain silent on how online communities can resolve other forms of tension. For example, when few members do not subscribe to the community's norms and values, they may engage in destructive knowledge flows such as vandalizing knowledge artifacts and harassing other members (Cheng et al. 2015).

Are organizational structure and participation dynamics sufficient to resolve tensions and sustain knowledge flows in online communities? In answering this question, we note that both perspectives subscribe to viewing the online community as an emergent form of organization. Community structure and participation dynamics emerge over time from repeated interactions among members over time. Against this common understanding of online communities, there is increased evidence that successful online communities have formal management that makes top-down decisions. Most of this evidence comes from the practitioner literature (Young 2013), but scholarly work is following suit. For example, studying an online healthcare community, Barrett et al. (2016) found that the success of the community resulted from the persistent engagement of multiple stakeholders including community founders, managers and members. Related, online community manager is a growing job category (Kerr and Kelleher 2015), and some scholarly work suggests guidelines and best practices for *designing* online communities (Ren and Kraut 2014).

To summarize, while the prior work suggests online communities deal with tensions via bottom-up participation dynamics and emergent organizational structure, the recent work suggests that active management, policies, and interventions are needed for resolving tensions and sustaining high-quality knowledge flows. How can top-down community management resolve tensions and sustain knowledge flows? Also, does top-down management complement the role of emergent structure and dynamics in resolving tensions or can a conflict occur between the top-down and bottom-up processes?

Methods

Our interest is to understand how an online community can be successfully managed and still retain the benefits associated with open boundaries, fluidity, and emergent sociality (Faraj et al. 2016; Faraj and Johnson 2011). Specifically, the phenomena of interest are tension resolution and sustaining knowledge-flows. Because these two phenomena involve the temporal interaction of multiple actors, they are better understood with an interactionist perspective (Kane et al. 2014). Our theoretical sampling strategy thus aimed at sampling the accounts of multiple actors including the managers, emergent leaders, and members in the online community. In this context, we deliberately select one online community that employs formal management. The context of healthcare suits well the purpose of the study. Healthcare is an institutionalized field that has strict expectations on how medical knowledge is shared and disseminated.

We study Mayo Clinic Connect (referred to as Connect hereafter), a leading online patient community sponsored by Mayo Clinic. The community is designed to "connect patients with each other and with those interested in learning more about Mayo Clinic or a particular health interest." ("Mayo Clinic Connect Community Guidelines" 2019). This community enables patients and their families with different medical backgrounds to share experiences, find support, and exchange information with others who have faced similar experiences. Important to our research context, the community has open boundaries. The community is open to the public and is not restricted to use by patients at Mayo Clinic. Currently the community has 85,000 registered members and 10,000+ active members who have made a post within the last 30 days.

We deliberately chose the context of healthcare, and online patient communities for three purposes. First, healthcare suits the purpose of the study well because it is an institutionalized field that has strict expectations on how medical knowledge is shared. Second, tensions are more rampant in healthcare communities because of the sensitivity of medical and personal information (McKee 2013) and the quality, accuracy and trustworthiness of medical information that is shared online (Benigeri and Pluye 2003; Marrie et al. 2013; Ogah and Wassersug 2013). Finally, as authors, we are generally interested in a healthcare context, and hope to identify ways to improve the well-being of patients through this research.

To familiarize us with the community, one author of this study participated in frequent informal discussions with the director of Connect and attended a community meeting discussing the directions, concerns, successes, and goals of the community. During the meeting, the author observed the interactions between the community management and members attending the meeting. Additionally, all authors joined the online community and observed members' contribution and interactions. The initial engagement with the community informed the theoretical sampling for the study.

The main source of data came from semi-structured interviews with key stakeholders in the community. The first semi-structured interview was with the director of the community. This interview was instrumental to guide the subsequent interviews through a snowballing approach (Bryant and Charmaz 2007). In two rounds of data collection, we interviewed the community director, the management staff, and fifteen other emergent leaders and members that were pointed to us in the initial interviews. At the time of the data collection the members we completed interviews with comprised of 100 percent formal management and 80 percent of the informal community leaders. Through probing questions, we asked about their participation in Connect, and what they felt the community did to achieve success. We asked them to expound on their experiences with the community – including challenging ones. Overall, we let the interviewees talk about their personal accounts of being member of Connect and interacting with others. Each interview lasted approximately 30 minutes and was digitally recorded and transcribed verbatim. At the end of each interview we asked the interviewee to suggest other people to talk to. We continued to interview key individuals until we reached theoretical saturation, which is defined as the state in which additional sampling and data analysis reveal no new concepts and no additional data are needed. (Glaser 2002).

To manage, code, and analyze our interview data, we utilized MaxQDA Analytics Pro. We adopted a grounded theory perspective in order to let theory emerge from the data. We followed a constant

comparative process in that throughout our analysis we repeatedly compared codes and findings in frequent discussion among the authors (Charmaz 2006). The process we took is demonstrated in the subsequent paragraphs and in Table 1.

Aspect of Study	**Methodological Considerations**	**Illustration from our study**
Data Collection	Initial Sampling	We were interested in the management of online communities. We selected a professionally managed online community in order to find a suitable case.
	Immersing into phenomenon	We became participants in the online community as well as attending a yearly training meeting between moderators and mentors
Concept Identification	Open Coding	We coded the interviews without any pre-defined themes, theories, or codes. The codes came from statements in the data. In total, we ended up with 295 codes.
	Second level coding	After reviewing the 295 codes, we combined those codes into higher-level abstractions. Creating these high-level categories of codes, we can establish relationships between groups
Relationship Identification	Axial Coding	According to Strauss and Corbin (1990), axial coding is done to relate categories to one another. We completed this by identifying relationships between practices and desired outcomes. We associated the member role implementing those practices, in what form, and why it was undertaken.
Theory construction	Connection to current theory	We follow the framework of Berente et al. (2019) in that the relationships we uncovered were then related back to our "pre-theoretical and theoretical understandings of an existing lexicon, namely the online community literature.
	Extension of theory	Previous theories in the online community literature suggest that tension resolution in online communities comes through socialization processes alone. After drawing insights from our data, we extend this current understanding with an enriched framework that includes top-down processes.

Table 1. Methodological Process

Two of the authors conducted line-by-line open coding techniques, as recommended by Strauss and Corbin (1990), on the first five interviews. After both authors completed the coding of these first five interviews, they discussed their findings to come to an agreement on what was emerging from the data. The remaining interviews were coded by only one of the authors, yet all authors met frequently during the coding process to discuss the inductive insights from the codes. After creating a large list of open codes, we conducted a second round of coding to ensure that minor differences in codes from open coding were combined to form unique ideas that did not overlap in the data. Overall, 295 unique relevant codes were identified that addressed our research question. Additionally, throughout this process of open coding, memos were kept on each interview to provide extended notes and insights. These additional observations in the memos drove our theorizing and discussions.

After completing open coding on all the interviews and discussing emergent themes we followed an axial coding process. This process in grounded theory is performed by combining and relating codes together to form more abstract themes and relationships (Corbin and Strauss 1990). Additionally, axial coding helps

to identify the who, what, when, where, why, and how of the emerging theory. In our case, it illustrated a clear picture of the top-down and bottom-up processes in the community.

Findings

The management team of Connect consists of a director and five staffers. These managers are referred to as moderators in the nomenclature of Connect. Because Connect is run by Mayo Clinic, it represents the values of its parent organization such as being safe, supportive, inclusive, and respectful ("Mayo Clinic Connect Community Guidelines" 2019). The community has formal guidelines to help members such as being careful when giving out medical advice, being respectful always, and no commercial posting or advertising. As the name moderator suggests, at the outset of the community, the moderators both ran the community, moderated others' posting, and provided the support to the members that started to join the community.

As the community grew, moderators recognized the need to scale the moderation of the community to maintain productive knowledge contribution. At the same time, the moderators noted that certain members were taking increasingly active leadership roles. They identified these emerging leaders and invited some of them to help monitor the community. In the nomenclature of Connect, the emerging leaders who were selected by the management are called mentors. Once selected, mentors, who volunteer their time, are trained and guided by moderators to consistently act in accordance with the guidelines of the community. Multiple training mechanisms are implemented in order to onboard and train mentors (Table 2).

Training Mechanism	Example from Interviews
Standard documentation	*"I'll send you three or four documents that you can look through and understand a little more what we're looking for in a mentor."*
Setting expectations	*"We're hopeful that you'll agree to spend at least X number of hours a month visiting the board and watching things and taking care of things."*
Coaching/Individual Training	*"When you are first invited, you spend time with the director, a one-to-one, doing screen shares and discussing different scenarios."*
Coaching/Group Training	*Moderators set up a private "mentors & moderators" group on Connect*
Mentor Meetups	"*The mentors and moderators, we get together once a year. That is fun to get to meet each other in person. It is a little bit more training and sort of continuing education of being a mentor."*
Remote Training	"*Every quarter we will have a phone call where we can share what concerns us and build up a discussion."*
Behavior Emulation	"*We [Moderators] ask them to emulate us as far as the way we would respond"*

Table 2. Training Mechanisms for Mentors

These training mechanisms communicate and transfer the goals of the community management (moderators) to the selected emerging leaders (mentors). This process brings three advantages. First, it reduces the burden of work load on moderators which is becoming infeasible with the continuous growth of the community, receiving around 10,000 posts per month. Second, as community members, mentors are uniquely qualified to assist members because of their shared experience – an experience that the moderators often do not have. Third, it enables moderators and mentors to specialize in different tasks. Moderators focus on resolving tensions, administrative responsibilities and setting up the agenda to improve community success and growth. Mentors can focus on patient success and support within the daily operation of the community. This specialization enables the community to deal with tensions when they arise.

Community Tensions

Online community tensions discussed in the literature surfaced in the interviews as well. Additionally, Connect is not only a successful online community, but also a healthcare community which creates further tensions around sharing medical advice, providing personal information, and reliance on community participation. We describe five prevalent situations where tensions arise. We provide a quote to demonstrate each situation. A more detailed table of evidence can be obtained by contacting the corresponding author.

First, many members are passionate about the community and their contributions to it. Many members experienced or overcame medical conditions and are passionate to share their experiences and help others improve their conditions. Passion can yield positive consequences such as increased engagement and participation. However, tensions may arise due to individuals arguing about differing beliefs and opinions. *"We might have a member that comes in really opinionated. I mean, I'm sure you've seen, in any online communication. Think about how ugly Facebook gets most of the time in the comments section. Some people bring that kind of behavior to the community" – Moderator 3*

Second, members in Connect are not required to disclose their real identities. These socially ambiguous identities allow individuals to share experiences without worrying about judgement or stereotyping and to alleviate privacy concerns. Yet negative consequences can come when identities are pseudonymous because anonymity creates an environment with no consequences. This increases the chance of social inhibition and therefore, the expression of anti-social behavior such as bullying, arguing, and trolling may emerge. *"Right, there're no names involved. It's just the member's handle and you can disclose or say what you want." – Mentor 8*

Third, sharing personal experience is perhaps the overarching activity of members in Connect. Members with similar problems share experiences to encourage, inform, and support one another. Overall, this activity creates positive outcomes and many members are very grateful for the feeling that they are "not alone." However, a tension arises when the experience is viewed as or claimed to be "medical advice". There is the potential for legal repercussions if that advice fails to provide positive outcomes or causes further harm. *"When it comes to alternative medicine, complementary medicine, that's where we run into problems. Because members are free to express their views, but when they advise their discussion group and they instruct them to stop taking traditional medications and go for homeopathy or herbal things, that's when we have to step in." – Moderator 4*

Fourth, soliciting personal information is helpful for community members to know how to respond to other members. More information regarding a member's condition and/or situation helps other members relate to their issues and provide relevant and necessary support from their knowledge and experience. Members often ask questions such as "how was this diagnosed", "how long have you had the problem", and "what kinds of treatment have you had." Though many times this is taken too far. Violations to information privacy can occur when a member shares too much personal health information. The community wants to ensure the privacy and safety of its members, and thus the proper amount of information provided is a unique tension to this health care community. *"In a HIPAA sense, this individual is asking this person very specific questions on the day, the time and the results of a medical test, and I'm not sure that that's an appropriate place for a public discussion board." – Mentor 6*

Fifth, reliance on the community is the extent to which a member utilizes the community to take care of their medical needs and questions. Thus, an increase in reliance is positive for the community because it signifies members will participate more. The community does its best to provide support and help to individuals. However, this can create a tension because the community is not a replacement for proper medical care, and the over reliance on the community might lead members to not go to the doctor, leading to negative consequences for their health. *"You can't really make a critical decision on somebody from a message board, so that should be obvious. If it's not, you point it out and just say, listen. We're just other members in a community who have had similar experiences, but you need to find a doctor." – Mentor 13*

Tension Resolution Practices

In the verbiage of the interviews, moderators and mentors described success as desired outcomes. Desired outcomes, from a tension perspective, can be interpreted as the positive consequences of resolving tensions. Two distinct categories of desired outcomes were identified from the codes – namely community outcomes and member outcomes. Community outcomes describe success at an organizational level and include positive culture, active engagement, and adherence to guidelines. Member outcomes focus on a member's well-being, support, and reactivation. We found that moderators and mentors enacted different tension resolution practices to achieve different desired outcomes. Figure 1 highlights these practices and outcomes.

In figure 1, the relationships between tension resolution practices and outcomes are the result of axial coding. Outcomes are displayed in the center with practices on the sides. We separated outcomes into patient success outcomes and community success outcomes. The labels of in-post and out-of-post practices signify in what domain a practice was enacted. Specifically, we found that some practices occurred inside the community discussion threads, and some occurred outside discussion threads. Therefore, we display in-post practices on the left side of the figure and out-of-post practices on the right. Groups of similar practices were abstracted into high-level categories that added parsimony to our understanding of the relationships. The first-level codes of practices can be found in the individual white boxes and the higher-level groups are distinguished by a gray box surrounding the first-level practices. These practices were not shared equally between moderators and mentors. We display the practice-outcome relationship that the majority of moderators implement in red, and the relationship that the majority of mentors implement in blue. This figure shows how moderators and mentors played different roles in addressing tensions of the community.

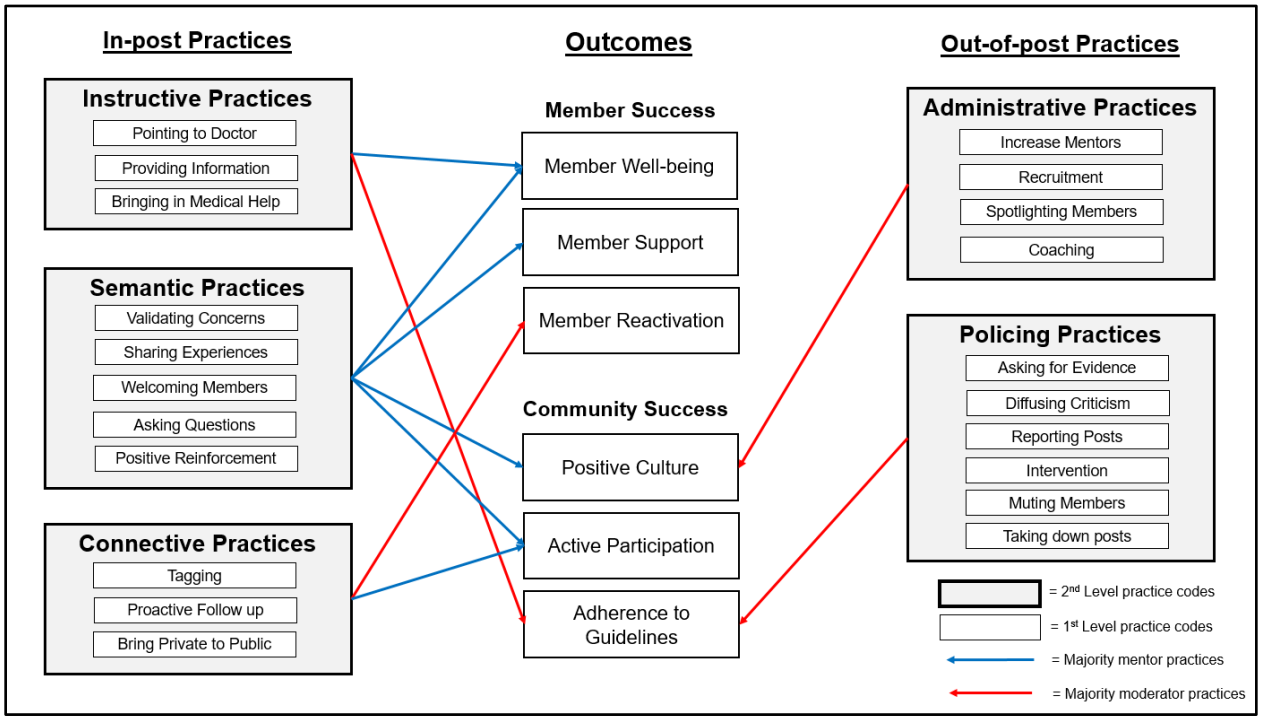

Figure 1. Groups of Tension Resolution Practices and Outcomes

The figure shows five tension resolution practice groups. Instructive, semantic, connective, administrative, and policing. Instructive practices are actions that provide medical information relevant to member experiences. For example, outside of discussion threads, moderators sometimes bring in medical help through web conferences and blog posts. Inside posts, however, Connect does not provide sanctioned medical advice. Providing information is solely through informing members of personal treatments and drugs used and sending

links to pages that could be beneficial. Lastly, pointing to doctor is the practice of encouraging members to seek professional help.

Semantic practices are actions that focus on the content of messages and responses, that is, what is being communicated inside a post. For example, validating concerns is the practice of reading a member post and relating empathically to what he/she is going through. Members share experiences when they include former experiences in posts. Welcoming members and asking questions are defined by their name. Positive reinforcement is including constructive and uplifting language in posts.

Connective practices are actions that create relationships inside the community and bring individual members into a community setting. Tagging members is accomplished with an "@" symbol as in many other social platforms. Proactive follow up is the practice of community leaders reaching out to individuals who have not participated recently. Bringing patients from private to public takes conversations that were initiated offline and inserts them into online discussion threads.

Administrative practices are performed by formal management outside of discussion threads to keep the community growing with a "positive community feel." This practice group contains a more diverse set of practices, but includes recruiting non-members to register and join the community through advertisements and marketing, the selection of mentors from emerging leaders in the member pool, spotlighting specific members in blog posts on the site, and coaching members through training mechanisms listed in Table 2.

Policing practices act as interventions against anti-social behaviors and departure from community guidelines. Sometimes policing practices can be inside of a post such as asking for medical evidence to back up a claim and diffusing criticism through language. Other practices happen outside discussion threads such as moderators intervening with additional authority to stop anti-social behavior. Members can be muted so that they can no longer post to a discussion, and posts can be deleted or edited by moderators to no longer display negative content.

Tension Resolution Outcomes

The tension resolution practices are instrumental for promoting community success outcomes. The first community success outcome, positive culture, is listed in the guidelines of Connect but was also seen as a core theme in the interviews with moderators and mentors. While moderators attempt administrative programs such as get to know you campaigns of members to create a positive culture, mentors contribute more to culture through semantic practices inside posts. Ultimately, the hope of management is that positive culture is achieved through members exemplifying mentors' behavior due to the social norms of reciprocity.

Mentor 6: When I see a new person come into the Connect community and the discussion area that I'm watching, I have a tendency to welcome them and try and establish some kind of early sense of the fact that we're a community, not just a cold chat board.

Mentor 5: When we do that and let them know we care they respond and then other members in turn have picked that up. Picked up on that practice and are now saying, "Oh, that's wonderful. I'm so glad to hear from you."

Second, active engagement in the community is the proactive and consistent participation from members in discussions. Engagement was achieved through two main practice categories: semantic practices and connective practices. While mentors implemented the majority of semantic practices, both moderators and mentors use connective practices. One explanation for why moderators continue to focus on connecting members is because they have been part of the community longer, and therefore have more knowledge of the relationships that could be connected. In contrast, semantic practices do not require knowledge of relationships. These practices only require taking interest in new posts from members – making mentors more able to initially engage in this type of practice.

Mentor 7: Asking questions of the person who is posting is a good thing. I know that people don't always like to post on a public forum, even if it's anonymous. I always start a sentence with if you are comfortable

sharing more information about yourself, would you please tell us how this was diagnosed and what kind of treatments you have experienced, et cetera?

Moderator 3: I try to connect them. A lot, being a moderator of the group, you learn the conversations that, you're familiar with the conversations, so sometimes, somebody will post something, and I'll go, "Oh! She talked about this, too!"

Third, adherence to guidelines entails the community not deviating from the posted guidelines on the website that include the community remaining "safe, supportive, inclusive, and respectful." The guideline referenced most in the interviews was being careful about giving out medical advice. The community needs to ensure that no member posts a personal solution to a disease or diagnosis without stating that this is just their personal experience. Additionally, the community watches for anti-social behaviors such as bullying, obscene language, and personal commercial advertisements. These types of transgressions are not permitted and outlined in the Terms of Use. These anti-social behaviors are addressed mainly by moderators using policing practices. Mentors do not participate directly in practices that enforced adherence to guidelines. Instead, mentors acted more in a reporting role, to report to moderators any evidence of anti-social behaviors contrary to the guidelines.

Moderator 2: There are times where that is a concern, where there might just be bad advice. But again, that's where we step in, in those situations.

Moderator 1: It's being there to intervene if people post their personal contact information, which could put them at some sort of privacy risk or they're writing in that they feel like they're close to ending their life or this kind of thing.

Mentor 10: I have asked a moderator to get involved on a couple of occasions where I thought someone had posted something that made me concerned.

In addition to achieving community success, the overarching goal for Connect is member success. The first member success outcome is member well-being, or a members' positive and safe progress with his/her personal health. This outcome is achieved through two practice groups: semantic and instructive practices. Most of these practices come from mentors because of their similar experiences in the past. Moderators, however, have the authority to speak to doctors and get some feedback, as well as periodically hosting webinars with doctors discussing common concerns and questions that have been raised by the community. These webinars are the only out-of-post practice in this category, where most of the practices to improve member well-being come from in-post.

Mentor 4: They soon come to realize that as we work with them, that what we're most anxious to do, is encourage them to take on more of the responsibility managing their own circumstance and dealing with their medical team.

Moderator 3: It's a live video Q&A session between a pediatric physician and the surgeon, or the physician. The two of them have conversation, we post that on Facebook and then we also have it linked to Mayo Clinic Connect, so the members there are able to watch that video live, ask questions before and during the live segment takes place.

Second, patient support is realized when a patient does not feel alone, has appropriate resources to meet their needs, and has the emotional support to traverse a difficult battle with whatever condition is affecting them. Based on the interviews, in-post semantic practices provide the most support for this outcome. Specifically, welcoming members, sharing personal experiences, and validating concerns was the content that provided patients with support. There is evidence of moderators using these practices, yet it is most often to give the mentors something to emulate. More recently, mentors are the main role focusing on this outcome due to the mentor's shared experience and the ability to relate to patients. An excerpt that highlights the roles of moderator and mentor is a moderator who talked about supporting a patient but clearly stated in the interview that she "took off her moderator hat" and acted as a mentor for that specific post.

Moderator 1: I don't very often take off my moderator hat and speak from a personal perspective, but someone wrote in about a situation they had and I had quite a similar one, and no one else was really responding, so I could provide some value and benefit to this member, really by kind of reinforcing that what she had was a valid problem, that kind of thing and explaining where things went in my situation.

Mentor 11: Yeah, so you have your peer-to-peer support, and people who have had the same experiences as you.

Third and finally, individual patient reactivation is a large focus for Connect. Many of the discussion groups revolve around chronic conditions, and therefore if a member with a health condition drops off the community and ceases to post, the community is not able to help and support that member. In order to achieve the outcome of patient reactivation, the moderators and mentors would tag inactive members and proactively follow up to see how they were doing and invite them to be an active participant. A relationship is needed before the mentor or moderator can follow up. Even though this appears to be more of a mentor task, moderators often follow up because of their existing relationships.

Mentor 7: I would go back and find people that I knew had been on before and weren't well and we just hadn't heard from them in a while. I would check back and ask them how they're doing.

Moderator 4: If a member has been posting and then suddenly stops, we tag members to connect them to our members

Discussion

Multiple tension resolution practices are used by Connect moderators and mentors to achieve the desired outcomes of community success and member success. These tension resolution practices are employed by the community management (moderators) and emerging leaders (mentors). These practices are not solely emergent nor completely prescribed. The nature of the practice matches the needs of the community and its members. Some of these practices are *interventive*. They are mostly used to provide a quick and direct resolution to a time-sensitive tension. Other practices are *preventive*. They promote members' behavior that lead to desired outcomes by avoiding the tension. We refer to these two higher-level groups of practices as *moderation* and *socialization* respectively.

Moderation practices respond to unfortunate events quickly and directly when lengthy deliberation is not an option. Such a swift reaction is needed to avoid negative consequences. Policing practices are the primary example of moderation practices. The additional authority that moderators have due to their formal role facilitated the ease of policing inside of the community. Below are some examples of negative consequences of tensions that moderation was specifically able to address.

First, passion is great for engaged individuals to share what they know and love, but when viewpoints are different there is a possibility for hurtful words and abrasive actions. In addition to passion, the negative effects of socially ambiguous identities create an environment that allows anti-social behavior to be posted. Finally, providing personal information can lead to a HIPAA violation if it is not addressed quickly. Moderation was used in all these scenarios through the practices of muting members, taking down posts, and even speaking to members through private messages. Where these moderation practices excel in comparison to socialization is in the fact that these practices could possibly be seen as unfair if a regular member enacted them, but due to the formal authority of moderators it is acceptable.

On the other hand, most tension resolution practices are not interventive but rather preventive. They serve to promote the good behavior and achieve desired outcomes through repeated interactions among members over time. Socialization was accomplished through the semantic practices (how content is constructed), instructive practices (sharing medical experiences and advice), and connective practices (establishing relationships with members). Leaders who implement semantic practices ask questions, inviting others to share experiences, and create a positive welcoming atmosphere where members feel comfortable sharing.

Semantic practices are influential in establishing positive social norms and bringing out the positive consequences of tensions.

For example, one tension that socialization practices can address is the patient's reliance on the community. We specifically saw this through the instructive practice of pointing a patient to their doctor and prompting patients to go and receive medical advice. To leverage the positive consequences of relying on a community, socialization connective practices, such as tagging community members and establishing relationships can help provide patients with the support they need as they rely on the community.

Through moderation and socialization, the community management and emerging leaders provide responsive resolutions to rising tensions. To sustain productive knowledge flows, the community management implements formal moderation practices in addition to relying on socialization practices that have been highlighted in previous literature (Faraj et al. 2011). The socialization of members has been shown in multiple studies to help decrease tensions (Faraj et al. 2011; Kane et al. 2014). However, the simultaneous existence of these bottom-up and top-down processes poses an important question. How can both moderation and socialization practices work together? Does moderation discourage free flowing ideas, and does socialization create unwieldy social norms that don't align with a community's goals? Both groups of practices can have a positive impact on the community when they are designed to leverage the positive aspects of both top-down moderation and bottom-up socialization. Figure 2 outlines our process model of a virtuous cycle of sustaining knowledge flows.

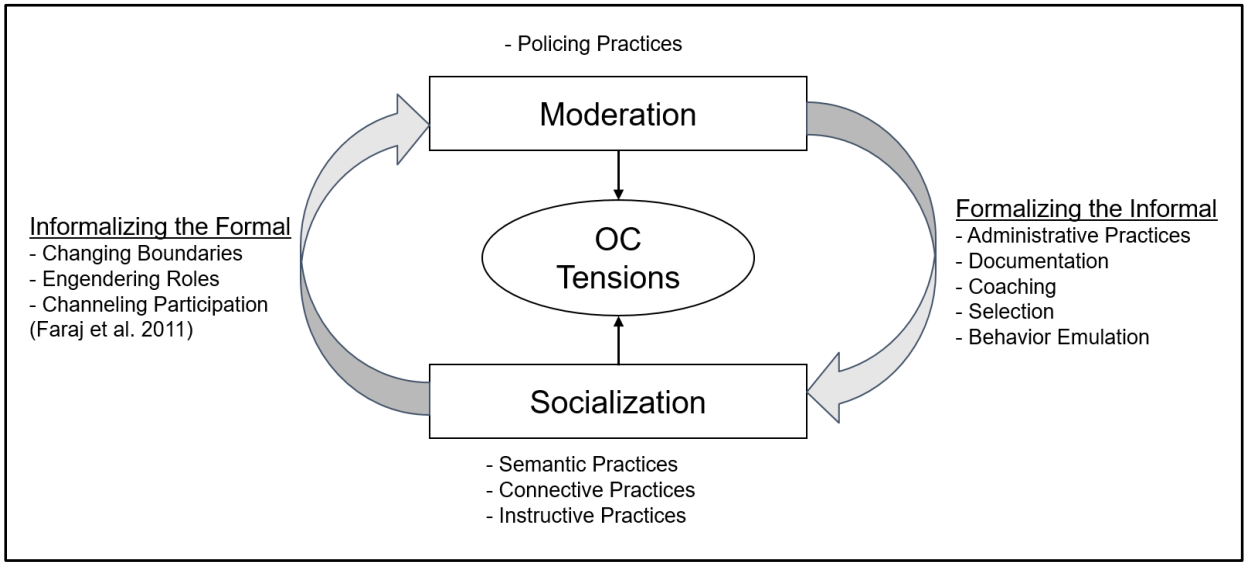

Figure 2. The Virtuous Cycle of Sustaining Knowledge Flows

The Virtuous Cycle of Formalizing the Informal and Informalizing the Formal

Moderation allows the community to respond to critical and time-sensitive tensions. Moderation requires the establishment of formal control mechanisms (outlined in Connect Terms of Service). The institution of these mechanisms and the evidence of their execution by moderators shows that at times socialization is insufficient. However, community management cannot moderate all members' posting due to the growing membership. Instead, community management leverages the emerging leaders in the community to help. Yet, these emergent leaders are not endowed with the privileges to execute these formal control mechanisms. The role of the emergent leaders is to exemplify the community norms that the moderators hope to achieve. Through onboarding and training mechanisms (see Table 2) such as one-on-one coaching, the emergent leaders can spread social norms that are not engendered from the ground up, but that are fostered from the formal management. We refer to this process as *formalizing the informal*. This process decreases the need of interventive moderation practices through the promotion of preventive socialization practices.

"We've come to a point where initially the moderators, despite the mentors welcoming the members or patients, moderators would have to go in and welcome the new members. Now we've come to a point where the mentors jump in to welcome the new member." – Moderator 4

The moderators are there to guide conversation and offer up advice along the way, but really support the mentors to be in the group and be in the room constantly talking with everybody. – Moderator 3

On the other hand, as the community grows, it becomes important to adapt the guidelines and the formal control mechanisms to match the need to the community and its members. The community management understands that with organic growth comes the need to understand what members are looking for. The community management learns about these needs from being part of the conversation. In addition, through face-to-face meetups, private online conversation, and online group meetings, the emerging leaders serve as a conduit to channel the members' experiences and the community's emerging issues to its management. We refer to this process as *informalizing the formal* because it provides the community management valuable input to evolve the community's guidelines and formal moderation mechanisms.

"I'm not worried about diluting sense of community just because the different roles that we, as moderators, play. We are all integrated within Connect." - Moderator 2

"Our members are now supporting each other. They're not confronting each other. They're smoothing things over themselves. We're reaping the rewards of the hard work that we've done and now we're taking that and seeing how we're going to move forward and how we keep this going?" - Mentor 5

Ultimately, the community management understood that excessive moderation stifles the progress of the community. The management is wary to post or edit a large amount of comments in order to avoid one common ideology and deter individual beliefs. The bottom up process of emergence follows close to the ideas of Faraj et al. (2011) in that socialization may channel participation, change boundaries, and engender new roles. This was demonstrated in Connect with new mentors emerging from the community, bringing new ideas, experiences and personalities to a formal management role. The success of the community is the result of its moderation practices reflecting the emerging organic social norms. Yet, these social norms are heavily shaped with prescribed organizational goals within the institutionalized healthcare field. As the community director puts it: "by modeling desired behavior, tensions are largely avoided. There is by far more support and camaraderie that takes place than tension."

Implications: Sustaining Knowledge Flows in Online Communities

Moderation and socialization are essential to resolve tensions and sustain knowledge flows in an online community. Together, these practices create virtuous cycle. This emerging process theory answers our research question of how online communities sustain productive knowledge flows in the presence of tensions. The community formal management can influence the socialization of the online community. The community emerging leadership can inform the moderation practices. As the cycle continues, guidance from above and social processes from below continue to provide productive knowledge flows while reducing negative consequences of tension, the presence of tensions and the needs to intervene.

In response to our second research question, we note that in our case, participation dynamics and emergent structure are not sufficient to resolve tensions and sustain knowledge flows. Emergent structure is not sufficient for Connect, especially because tensions demand swift responses. We acknowledge that there may be scenarios and contexts where pro-social behavior is not as critical, and therefore over time, the community has time to self-correct and eventually generate sufficient social norms and leadership to address the tensions. However, the case presented in this paper provides a clear counter evidence of the utopic view of online communities as self-organizing emerging structures capable of creating and sustaining knowledge flows. Socialization processes are necessary but not sufficient in sustaining productive knowledge flows.

Second, the case supports the accumulating evidence of the prevalence of anti-social behavior in online communities. Such behavior compromises the benefits of knowledge flows and it is not clear how sociality

alone in absence of formal control mechanisms can counter this effect. Our study directly addresses this lack in understanding by demonstrating how communities, such as healthcare or organization-sponsored, could address tensions more efficiently by implementing formal moderation approaches to complement socialization. Such formal control does not negate the organic and emerging nature of the communities but rather refines it through connecting formal moderation and emergent socialization in a virtuous cycle. This new theoretical understanding helps us see online communities as organized emergent structures rather than completely emergent organizational structures.

The nature of our case study and research questions allows these findings to apply specifically to formally managed online communities. However, we believe that as an online community with emergent structure matures to a more formal extent, such as Wikipedia, the virtuous cycle would still apply if put into practice. While we understand this cycle may not be needed or even plausible for all online communities, we would advocate taking a long look at what a community's needs are to see if this model can help improve outcomes in an online world with ever increasing anti-social behavior and tensions. Pro-social requirements may not always be considered by online communities, yet we suggest that this consideration will help in designing practices and processes that can be put in place to achieve the goals of the organization. Hopefully these can be a guide and reference for new communities seeking productive ideas.

Conclusion and Future work

Increasingly, online communities are incorporating additional structure to moderate communities and little is known about which contexts this process will be beneficial. Through an inductive, grounded theory study of Mayo Clinic Connect, we have theorized a virtuous cycle to sustain productive knowledge flows in online communities experiencing tensions. As an extension to previous literature that concludes that emergent sociality in online communities is sufficient to address tensions, we argue that formal moderation and the process of formalizing the informal members of the community will benefit communities that require pro-social behaviors in order to succeed. Future research would benefit by looking at additional cases and additional contexts beyond healthcare to uncover when socialization is sufficient and when this process of socialization and moderation is needed.

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